

Xinyu Mao

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RESEARCH INTERESTS

My research focuses on the foundations of cryptography, including the construction and limitations of cryptographic primitives. I also study communication complexity and learning theory. My recent work explores pseudorandom codes motivated by robust watermarking of generative AI.

ACADEMIC APPOINTMENTS

University of Waterloo Starting Oct. 2026
Incoming Postdoctoral Fellow
Co-hosts: Mohammad Hajiabadi and Douglas Stebila

EDUCATION

University of Southern California Aug. 2022 – Aug. 2026
Ph.D. in Computer Science
Advisor: Jiapeng Zhang

Shanghai Jiao Tong University Aug. 2018 – Jun. 2022
Bachelor of Engineering in Computer Science
ACM Honor Class, Zhiyuan College; class rank: 2/33

PUBLICATIONS

Authors are listed alphabetically by surname.

- [1] Prediction Under Imperfect Compression: A Theory of Approximate MDL
Qian Li, **Xinyu Mao**, Shang-Hua Teng, and Guangxu Yang
NeurIPS 2026 — Oral; to appear
- [2] Finding Random Collisions for Random Degree-2 Functions
Xinyu Mao
TCC 2026 — to appear; sole author
An efficient collision sampler rules out distributional collision resistance for random degree-2 maps over finite fields in most parameter regimes.
- [3] Advancing Pseudorandom Codes: Beyond Parity Checks and Standard-Model CCA1 Security
Yu Chen, **Xinyu Mao**, and Hongxu Yi
ASIACRYPT 2026 — to appear
- [4] Sublinear Proofs over Polynomial Rings
Mi-Ying Miryam Huang, **Xinyu Mao**, and Jiapeng Zhang
ASIACRYPT 2026 — to appear
Introduces new techniques for constructing lattice-based succinct non-interactive argument of knowledge (SNARK).

PUBLICATIONS (CONTINUED)

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- [5] A Min-Entropy Approach to Multi-Party Communication Lower Bounds
Mi-Ying (Miryam) Huang, **Xinyu Mao**, Shuo Wang, Guangxu Yang, and Jiapeng Zhang
CCC 2025
 - [6] Universal Computational Extractors and Multi-Bit AIPO from Lattice Assumptions
Yilei Chen and **Xinyu Mao**
EUROCRYPT 2025
Introduces obviously programmable functions as a tool for constructing universal computational extractors and multi-bit auxiliary-input point obfuscation.
 - [7] Gadgetless Lifting Beats Round Elimination: Improved Lower Bounds for Pointer Chasing
Xinyu Mao, Guangxu Yang, and Jiapeng Zhang
ITCS 2025
 - [8] On the Power of SVD in the Stochastic Block Model
Xinyu Mao and Jiapeng Zhang
NeurIPS 2023
 - [9] Communication Lower Bounds of Key-Agreement Protocols via Density Increment Arguments
Mi-Ying Huang, **Xinyu Mao**, Guangxu Yang, and Jiapeng Zhang
TCC 2023
 - [10] Non-Adaptive Universal One-Way Hash Functions from Arbitrary One-Way Functions
Xinyu Mao, Noam Mazor, and Jiapeng Zhang
EUROCRYPT 2023

PREPRINTS

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- [P1] Black-Box Separation Between Multi-Collision Resistance and Collision Resistance
Xinyu Mao and Jiapeng Zhang
Preprint, 2025 — Cryptology ePrint Archive, 2025/2049

EARLIER RESEARCH EXPERIENCE

University of Southern California	May 2021 – Apr. 2022
Research Intern; advisor: Jiapeng Zhang	
Foundations of cryptography	
Shanghai Jiao Tong University	Sep. 2020 – Feb. 2021
Undergraduate Researcher; advisor: Shengli Liu	
Public-key cryptography	

TEACHING EXPERIENCE

University of Southern California

Teaching Assistant

Introduction to Online Optimization/Learning	Spring 2026
Quantum Computing and Quantum Cryptography	Fall 2025
Applied Cryptography	Spring 2025
Introduction to the Theory of Computation	Fall 2023

Shanghai Jiao Tong University

Teaching Assistant

Great Ideas in Computer Science	Fall 2019, 2020, 2021
Developed an Enigma machine simulator to support teaching.	

ACADEMIC SERVICE

Conference reviewing

CRYPTO	2025, 2026
TCC	2024, 2025
ITCS	2025
ICALP	2026
FOCS	2026
SODA	2026

HONORS AND AWARDS

Zhiyuan Outstanding Student Scholarship	2022
Shanghai Outstanding Graduate	2022
Foresight-Sequoia Scholarship	2021
Huawei Scholarship	2020
Zhiyuan Honorary Scholarship	2018, 2019, 2020
Bronze Medal, China National Olympiad in Informatics (NOI)	2017